	National University of Sciences and Technology
	Pakistan Navy Engineering College
	Department of Electronics and Power Engineering
	Quiz-1 (MS Artificial Intelligence – Fall 2026)

Course Code: CS-871 **Course Title:** Machine Learning
Instructors: Dr. Ihsan Ullah Khalil **Timing:** During Class Timing
Max. Marks: 3 **Max. Time: 20** minutes
Name: _____ **Regn No:** _____

S/No.		Marks
Q-1	An autonomous combat drone needs to predict the threat level (y) of an unidentified aircraft based on a feature vector x, which includes radar cross-section, speed, and emission frequencies. The targeting system compares this feature vector against a database of known past encounters, $x^1,\ldots,x^n,y^1,\ldots,y^n$. The system must predict the threat level using a weighted average of the known outcomes, where the influence of a past encounter drops off depending on its feature vector distance to the current target. The aerospace engineers also need a tunable dial to control how rapidly this distance-based influence decays. Which specific predictor should the engineers implement, and what parameter allows them to control how strictly the system relies on the most similar past encounters? Hint: $$w^i = \frac{e^{-\|x-x^i\|_2^2/\rho^2}}{e^{-\|x-x^1\|_2^2/\rho^2} + \cdots + e^{-\|x-x^n\|_2^2/\rho^2}}$$	
Answer	**Soft nearest neighbor predictor is to be used.** Explanation: Instead of selecting a single historical data point, the drone calculates its current threat prediction (y') as a combination of all known past outcomes (y^i). The influence of any historical encounter is determined by a calculated weight (w^i), which is strictly a function of its spatial proximity to the current target in the feature space**. The Numerator:** $e^{-\|x-x^i\|_2^2/\rho^2}$ The system computes the squared distance $\|x-x^i\|_2^2$ between the new target's feature vector x and a historical target's vector x^i. Because this distance acts as a negative exponent, it inverts the relationship: a spatial distance of zero results in an unscaled similarity score of 1 (e^0). As the mathematical difference between the aircraft (e.g., varying speeds or radar cross-sections) grows, the negative exponent increases, causing that specific encounter's influence to exponentially decay toward zero. **The denominator** computes the total sum of the exponential similarity scores across the entire database of n encounters: $e^{-\|x-x^1\|_2^2/\rho^2} + \cdots + e^{-\|x-x^n\|_2^2/\rho^2}$. Dividing a specific encounter's numerator by this global sum yields the normalized weight w^i. The system then calculates the final prediction y' by taking the sum of every historical outcome y^i multiplied by its respective weight w^i. For aerospace engineers calibrating the UCAV, the parameter $\rho > 0$ defines the characteristic length of the decay. This is why a small ρ forces the model to revert to a standard nearest neighbor predictor that only looks at the absolute closest points. Ultimately, the final equation $y' = \sum_{i=1}^{n} w^i y^i$ generates the prediction by multiplying every known outcome y^i by its newly calculated fractional weight w^i, seamlessly blending the historical data based on proximity.	